

GASTRIL 50MG/2ML INJECTION (2ML AMPOULE)

DESCRIPTION: A clear, very pale yellow liquid.

COMPOSITION: Each ml contains Ranitidine HCl equivalent to Ranitidine 25mg

PHARMACODYNAMICS:

Animal experiments both in vitro and in vivo have established that ranitidine is a selective, competitive antagonist of histamine at H₂-receptor sites. Ranitidine has no significant interaction at histamine H₁-receptors, muscarinic receptors or beta-adrenoceptors. Ranitidine is a potent inhibitor of gastric secretion in the rat and dog. All the evidence from human studies is compatible with a selective, competitive antagonism of histamine H₂-receptors by ranitidine in man. Oral or intravenous administration of ranitidine inhibits both basal gastric secretions and gastric acid secretion induced by histamine, pentagastrin and other secretagogues. On a weight basis ranitidine is between 4 and 9 times more potent than cimetidine.

After oral administration of ranitidine, the plasma concentrations of ranitidine achieved are directly related to the dose administered. A plasma ranitidine concentration of 50-100 mg/mL has an inhibitory effect upon stimulated gastric acid secretion of approximately 50%.

Inhibition of pentagastrin-induced gastric acid secretion increases with dose, being approximately 90% two hours after an oral 150 mg dose and a significant effect is still evident 12 hours after this dose. In 10 patients with duodenal ulcer, 150 mg ranitidine given orally every 12 hours significantly reduced mean 24 hour hydrogen ion activity by 69% and nocturnal gastric acid output by 90% whereas cimetidine (200 mg three times daily and 400 mg at night) reduced mean 24 hour hydrogen ion activity by 48% and nocturnal gastric acid output by 70%.

Pepsin secretion is also inhibited by ranitidine, but secretion of gastric mucus is not affected. Ranitidine does not alter the secretion of bicarbonate or enzymes from the pancreas in response to secretin and pancreozymin. Reduction in gastric acid secretion induced by ranitidine 150 mg twice daily for 7 days did not cause bacterial overgrowth in the stomach.

Pulse rate, blood pressure, electrocardiogram and electroencephalogram were not significantly affected in man following recommended doses of ranitidine.

Chronic ranitidine therapy (300 mg/day for 28 days) had no effect on serum prolactin, gastrin, thyroid stimulating hormone, follicle stimulating hormone, luteinising hormone, gonadotrophins, testosterone, oestriol, progesterone or cortisol levels.

PHARMACOKINETICS: Peak plasma levels occur about 2-3 hours after oral administration of ranitidine. Absorption is not significantly altered by food or concurrent antacid administration. Bioavailability of ranitidine is approximately 50%. Serum protein binding of ranitidine in man is in the range 10-19%. The elimination half-life is approximately 2 hours.

Ranitidine is excreted via the kidneys mainly as unchanged drug and in minor amounts as the N-oxide, S-oxide and desmethyl metabolites. The 24 hour urinary recovery of free ranitidine and its metabolites is about 40% after oral administration of the drug. Impairment of renal function requires a reduction in dosage (see Warnings and Precautions). Impairment of liver function may increase the bioavailability of ranitidine but has no significant effect on the elimination half-life. However, in the presence of normal renal function, no dosage reduction for oral or intravenous ranitidine appears necessary in patients with hepatic impairment. **Special Patient Populations**

Patients over 50 years of age: In patients over 50 years of age, half-life is prolonged (3-4 h) and clearance is reduced, consistent with the age-related decline of renal function. However, systemic exposure and accumulation are 50% higher. This difference exceeds the effect of declining renal function, and indicates increased bioavailability in older patients.

INDICATIONS: Indicated for the Treatment of duodenal ulcer, benign gastric ulcer, post-operative ulcer, reflux oesophagitis, Zollinger-Ellison syndrome, Prophylaxis of stress ulceration in seriously ill patients, prophylaxis of recurrent hemorrhage from peptic ulcer and prophylaxis of Mendelson's syndrome.

RECOMMENDED DOSAGE:

Ranitidine Injection may be given either as a slow (over a period of at least two minutes) intravenous injection of 50mg, after dilution to a volume of 20ml per 50 mg dose, which may be repeated every six to eight hours; or as an intermittent intravenous infusion at a rate of 25mg per hour for two hours; the infusion may be repeated at six to eight hour intervals, or as an intramuscular injection of 50mg (2ml) every six to eight hours.

Accumulation of ranitidine resulting elevated plasma concentrations will occur in patients with renal impairment (creatinine clearance less than 50 ml/min). It is recommended that ranitidine injection be administered in doses of 25mg.

Prophylaxis of Mendelson's Syndrome (prophylaxis of aspiration pneumonitis): Intramuscular or slow intravenous injection, 50mg administered forty five to sixty minutes before induction of general anesthesia.

Prophylaxis of hemorrhage from stress ulceration in seriously ill patients or prophylaxis of recurrent hemorrhage in patient bleeding from peptic ulceration parenteral administration may be continued until oral feeding commences. Patients considered to be still at risk may then be treated with ranitidine tablets 150 mg twice daily.

In the prophylaxis of upper gastrointestinal hemorrhage from stress ulceration in seriously ill patients, a priming dose of 50mg as a slow i.v. injection followed by a continuous i.v. infusion of 0.125 – 0.250 mg/kg/h may be preferred.

Prophylaxis of stress ulceration in seriously ill patients

Patients over 50 years of age (see Pharmacokinetics, Special Patient Populations, Patients over 50 years of age)

Renal Impairment

Accumulation of ranitidine with resulting elevated plasma concentrations will occur in patients with renal impairment (creatinine clearance less than 50 ml/min). It is recommended that ranitidine injection be administered in doses of 25mg.

ROUTE OF ADMINISTRATIONS: For i.m or slow i.v injection

CONTRAINDICATIONS: Patients with known hypersensitivity to Ranitidine.

WARNING & PRECAUTIONS:

Gastric ulcer: Treatment with a histamine H₂-antagonist may mask symptoms associated with carcinoma of the stomach and therefore may delay diagnosis of the condition. Accordingly, where gastric ulcer is suspected, the possibility of malignancy should be excluded before therapy with Ranitidine Syrup, Tablets or Injection is instituted.

Use in impaired renal function: Ranitidine is excreted via the kidney and in the presence of severe renal impairment plasma levels of ranitidine are increased and prolonged. Accordingly in the presence of significant renal impairment, serum levels should be monitored and dosage adjustments made. The clearance of ranitidine is increased during haemodialysis.

Children: Experience with ranitidine tablets in children is limited and such use has not been fully evaluated in clinical studies. It has however, been used successfully in children aged 8-18 years in doses up to 150 mg twice daily.

Long-term use: The risk of ulcer recurrence is determined by many factors. In some cases, long periods of treatment may be necessary and/or repeated. Evidence from controlled clinical trials of up to 18 months continuous treatment with Ranitidine has not revealed any undue untoward effects.

Bradycardia: In association with rapid administration of Ranitidine Injection has been reported rarely, usually in patients with factors predisposing to cardiac rhythm disturbances. Recommended rates of administration should not be exceeded.

Higher doses: The use of higher than recommended doses of intravenous H₂-antagonists has been associated with rises in liver enzymes when treatment has been extended beyond five days.

Porphyria: Rare clinical reports suggest that ranitidine may precipitate acute porphyric attacks. Ranitidine should therefore be avoided in patients with a history of acute porphyria.

Gastric pH: Agents that elevate gastric pH may increase the already present risk of nosocomial pneumonia in intubated ICU patients receiving mechanical ventilation.

INTERACTION WITH OTHER MEDICATIONS:

Ranitidine has the potential to affect the absorption, metabolism or renal excretion of other drugs. The altered pharmacokinetics may necessitate dosage adjustment of the affected drug or discontinuation of treatment.

Interactions occur by several mechanisms including:

1) Inhibitions of cytochrome P450-linked mixed function oxygenase system: Ranitidine at usual therapeutic doses does not potentiate the actions of drugs which are inactivated by this enzyme system such as diazepam, lidocaine, phenytoin, propranolol and theophylline. There have been reports of altered prothrombin time with coumarin anticoagulants (e.g. warfarin). Due to the narrow therapeutic index, close monitoring of increased or decreased prothrombin time is recommended during concurrent treatment with ranitidine.

2) Competition for renal tubular secretion:

Since ranitidine is partially eliminated by cationic system, it may affect the clearance of other drugs eliminated by this route. High doses of ranitidine (e.g. such as those used in treatment of Zollinger-Ellison syndrome) may reduce the excretion of procainamide and N-cetylprocainamide resulting in increased plasma levels of these drugs.

3) Alteration of gastric pH:

The bioavailability of certain drugs may be affected. This can result in either an increase in absorption (e.g. triazolam, midazolam, glipizide) or a decrease in absorption (e.g. ketoconazole, atazanavir, delavirdine, gefitinib)

USE IN PREGNANCY & LACTATION: **Use in pregnancy:** Ranitidine crosses the placenta. Adequate and well-controlled studies in humans have not been done. Reproduction studies performed in rats and rabbits have revealed no evidence of impaired fertility or harm to the foetus due to ranitidine. If the administration of Ranitidine is considered to be necessary, its use requires that the potential benefits be weighed against possible hazards to the patient and to the foetus.

Use in lactation: Breast feeding not recommended because ranitidine is distributed into breast milk and would possibly suppress gastric acidity, inhibit drug metabolism, and cause central nervous system (CNS) stimulation in the nursing infants.

SIDE EFFECTS: The following have been reported as events in clinical trials or in the routine management of patients treated with ranitidine. The relationship to ranitidine therapy has not been clear in many cases. Headache, sometimes severe, has been reported in a very small proportion of patients.

Central Nervous System: Rarely, malaise, dizziness, somnolence, insomnia and vertigo. Rare cases of reversible mental confusion, depression and hallucinations have been reported, predominantly in severely ill and elderly patients. In addition, reversible involuntary movement disorders have been reported rarely. There have been a few reports of reversible blurred vision suggestive of a change in accommodation. Reversible impotence has been reported rarely.

Cardiovascular: As with other H₂-receptor antagonists rare reports of tachycardia, bradycardia, premature ventricular beats, AV block, and asystole.

Gastrointestinal: Constipation, diarrhoea, nausea/vomiting, abdominal discomfort/pain.

Hepatic: There have been occasional reports of hepatitis, hepatocellular or hepatocanalicular or mixed, with or without jaundice. These were usually reversible.

Musculoskeletal: Rare reports of arthralgias and myalgia.

Haematologic: Rare reports of agranulocytosis or pancytopenia, sometimes with marrow hypoplasia or aplasia have been reported. Blood count changes (leucopenia, thrombocytopenia) have occurred in a few patients. These are usually reversible.

Endocrine: Controlled studies in animals and man have shown no stimulation of any pituitary hormone by ranitidine, no anti-androgenic activity, and cimetidine-induced gynaecomastia and impotence in hypersecretory patients have resolved when ranitidine was substituted. However, occasional cases of gynaecomastia, impotence and loss of libido have been reported in male patients receiving ranitidine but the incidence did not differ from that in the general population. **Integumental:** Rash including rare cases of mild erythema multiforme. Rare cases of vasculitis and alopecia have been reported.

Other: Rare cases of hypersensitivity reactions (eg, fever, bronchospasm, anaphylactic shock, rash, eosinophilia), small increases in serum creatinine. Acute pancreatitis has been reported rarely.

SYMPTOMS AND TREATMENT OF OVERDOSE: There has been virtually no experience with overdosage with Ranitidine Injection and limited experience with oral doses of ranitidine. Reported acute ingestions of up to 18 g orally have been associated with transient adverse effects similar to those encountered in normal clinical experience. (See Adverse Reactions). Symptomatic and supportive therapy should be given as appropriate. If need be, the drug may be removed from the plasma by haemodialysis. Rapid bolus injection of 300 mg intravenously (six times the recommended dose which should be given slowly) caused dizziness and peripheral vasodilatation

INCOMPATIBILITIES : Gastril Injection has been shown to be compatible with the following intravenous infusion fluids:- Normal saline, Dextrose 5% in water, Dextrose 10% in water, Ringer's lactate injection and sodium bicarbonate 5% injection. All unused admixtures of Gastril Injection with Infusion fluids should be discarded 24 hours after preparation.

STORAGE CONDITIONS: Store below 30°C. Protect from light. Keep out of reach of children. Jauhkan daripada kanak-kanak. Injection should not be used if discoloured or if a precipitate is present.

PACK SIZE: Pack in box of 10 x 2ml ampoule

SHELF LIFE: Please refer to the outer package

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

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